

Ideas Parked — Not Yet Merged (v13)

Staging document; each entry marks its eventual destination

This version parks a single, connected question: what physics-flavored names, if any, the atom-level quantities of Axiom 7 should carry. It is prompted by three things in one conversation — a proposed atomic-number/mass-number reading of N and a new count D , a proposed “information density,” and a proposed use of density for inertia in Assembly — and by what checking each against the current paper turned up: two physics names were already tried here once and deliberately retired, and an inertia-like weighting was already tested directly in Assembly’s own motion equation and found to break it. Nothing below is adopted. The paper is left exactly as `PA_ReWrite_V15_27.tex` has it; this file is the weighing the next version, if one is written, should start from.

1 The full inventory, exactly as it now stands

Seven scalar metrics follow from one Collapsed atom’s N vectors. Quoted precisely, not reconstructed from memory:

Quantity	Formula	Current name
S_{atom}	$\sum_i r_i$	atom strength
$\mathcal{V}_{\text{atom}}$	$\sum_i r_i^3$	atom volume
γ_{atom}	$\mathcal{V}_{\text{atom}}/N$	mean vector volume
$\mathcal{E}_{\text{atom}}$	$\sum_i r_i^2$	informational (intrinsic) energy
α_{atom}	$\sum_{i<j} r_i r_j (\hat{n}_i \cdot \hat{n}_j) / \sum_{i<j} r_i r_j$	atom coherence
σ_{atom}	$\frac{N^2}{N-1} \text{Var}(s_1, \dots, s_N), s_i = r_i^3 / \mathcal{V}_{\text{atom}}$	volume concentration
$\sigma_{\text{shell}}(w)$	shell-binned version of the same idea	shell concentration

Three of these once carried different names, retired by the paper’s own earlier draft, in the paper’s own words: “*Earlier drafts called $\mathcal{V}_{\text{atom}}$ ‘mass,’ γ_{atom} ‘density,’ and S_{atom} ‘atom energy’ — all three retired as misleading: the first two because neither participates in momentum, kinetic energy, or any calculation the word ‘mass’ would ordinarily license, and the third because a genuine notion of energy exists below and is not S_{atom} .*” That genuine notion is $\mathcal{E}_{\text{atom}}$, and its name was *kept*, not retired — the one case where a physics word survived contact with the paper’s own scrutiny.

Why $\mathcal{E}_{\text{atom}}$ ’s name survived and the other three did not. This is the load-bearing fact for everything below. $\mathcal{E}_{\text{atom}} = \sum r_i^2$ is not merely called energy because it resembles one; it is tied to another, independently defined quantity by an exact, verified identity:

$$|V_{\text{derived}}|^2 = \mathcal{E}_{\text{atom}} + 2 \sum_{i<j} r_i r_j (\hat{n}_i \cdot \hat{n}_j),$$

matching $\text{Var}(\sum X_i) = \sum \text{Var}(X_i) + 2 \sum_{i<j} \text{Cov}(X_i, X_j)$ term for term, verified to six decimal places. $\mathcal{V}_{\text{atom}}$, γ_{atom} , and S_{atom} -as-energy had no such identity; they were evocative without being

earned, and the paper’s own retraction says so almost in those words (“neither participates in . . . any calculation the word would ordinarily license”).

The rule this suggests, stated plainly. A physics name for a quantity in this paper should be kept only when the quantity is shown to participate in something provable — an exact identity or a forced structural correspondence to another already-established quantity — not merely when it resembles, in isolation, what a physicist or chemist would call by that name. This is not a new rule invented for this entry; it is the rule the paper has already applied once, the hard way, and this entry only makes it explicit so the next naming decision does not have to relearn it.

2 N and D : the mass-number / atomic-number reading, weighed against that rule

The most recent merged version of the paper (PA.ReWrite_V15.27.tex, Process principle: Collapse) already states, as settled content, a count $D = |A_1 \uplus \dots \uplus A_n|$, the number of value-distinct vectors among a Collapsed atom’s N , with $D \leq N$ exact and $N - D$ precisely the repeat count. This is offered there under a hedged chemistry analogy — N as mass number, D as atomic number — with the hedge that the analogy “should not be pushed further... a neutron is a genuinely different particle from a proton, not a repeated one.”

Checked against the rule above: this one earns it. Unlike the retired $S_{\text{atom}}/\mathcal{V}_{\text{atom}}/\gamma_{\text{atom}}$ names, D is not merely evocative. It is forced to equal $|A_1 \uplus \dots \uplus A_n|$ by $\uplus$ ’s own already-proven idempotence ($A \uplus A = A$) — an exact identity connecting a Collapse-side quantity to a Bundle-side one, the same shape of justification that kept $\mathcal{E}_{\text{atom}}$ ’s name. $N = D + (N - D)$ is also exact, mirroring $A = Z + (\text{neutrons})$ structurally, not just by resemblance. On the standard stated above, N/D ’s analogy names have a stronger claim to survive than S_{atom} , $\mathcal{V}_{\text{atom}}$, or γ_{atom} ever did, and about as strong a claim as $\mathcal{E}_{\text{atom}}$ ’s own.

Where the analogy still falls short of $\mathcal{E}_{\text{atom}}$ ’s. $\mathcal{E}_{\text{atom}}$ ’s identity connects it to a quantity used elsewhere in the algebra (V_{derived} , which feeds T ’s higher-dimensional actor). D ’s identity connects it to Bundle’s count, which is correct and exact but is not yet shown to feed anything further — no computation in the paper currently reads D off a Collapsed atom the way T reads V_{derived} . The name is earned in the narrow, structural sense; it has not yet been shown useful in the broader sense $\mathcal{E}_{\text{atom}}$ has. Worth tracking as a distinction, not a disqualification.

3 “Information density”: why it does not yet pick out anything

Proposed this round: an information density, “the sum total of centers and vectors in an atom,” possibly weighted by total strength alone. Checked against what the phrase could mean:

- If it means N (total count): already named, already has an analogy (mass number), adds nothing new under a new word.
- If it means $D + (N - D)$: this is N again, exactly, by arithmetic; same conclusion.
- If it means $S_{\text{atom}} = \sum r_i$ (“weighted by total strength”): already named (atom strength), and its one-time “energy” name is the specific one the paper’s own retraction explains, since the genuine energy quantity is $\mathcal{E}_{\text{atom}}$ and is not S_{atom} .

- If it means S_{atom}/N (mean strength, uncubed): this is the one candidate not already named anywhere in the current inventory. It is γ_{atom} ’s un-cubed sibling — where $\gamma_{\text{atom}} = \mathcal{V}_{\text{atom}}/N$ averages the cubed, volume-flavored quantity, S_{atom}/N would average the raw one. Not yet shown to participate in anything the way $\mathcal{E}_{\text{atom}}$ or D do; a candidate for future scrutiny, not a name ready to adopt.

Why “density” specifically should stay retired regardless of which formula is meant.

The word was already tried, for γ_{atom} , and retired by the paper’s own earlier draft for the same reason mass was: it does not participate in anything a physical density would license. Reusing the same retired word for a different formula does not un-retire it; it repeats the mistake under new cover. If S_{atom}/N turns out, on future scrutiny, to earn a name by the rule in Section 1, it should earn its own, not the one already spent.

4 Inertia in Assembly: closed, not open

This is the one item in this entry that is not a naming question at all, and should not be filed as one. The current paper states, directly, in the Enclosure’s own motion section: *“There is no mass term and no momentum carried between steps. This is deliberate rather than an omission. A unit carries no inertia, so it cannot overshoot... Weighting each step by a mass-like quantity was tested and produces oscillation rather than rest, precisely where the weighting is most extreme — a ninefold difference in strength gives a 729-fold weighting under a cubed reading, and a demonstrably non-monotonic trajectory.”*

$9^3 = 729$ is exactly the cubed-strength weighting any “density”-as- $\mathcal{V}_{\text{atom}}$ -flavored quantity would reintroduce if used to weight Assembly’s motion step. This has already been run, not merely anticipated, and the result was a specific, named failure (oscillation against a proven monotonic baseline), not an inconclusive one. Proposing a density metric for future use in Assembly inertia should not be filed as an open question needing a decision; it should be filed as a question this project has already asked and already answered, in the negative, with a number attached. Any future descriptive density-like metric is welcome to exist the way S_{atom} , $\mathcal{V}_{\text{atom}}$, and γ_{atom} already do — as a pure descriptive quantity that does not feed Motion — but not as a candidate for inertia specifically.

5 Weighing the options, and a recommendation

Combinations actually available, not an exhaustive list of words:

Option	Assessment
Keep N/D as mass number/atomic number, as currently merged	Earns its name by the Section 1 rule; the weaker of the two justifications ($\mathcal{E}_{\text{atom}}$'s own use elsewhere is stronger than D 's, which is proven but not yet used downstream). Lowest-risk of the live options.
Retire the N/D chemistry analogy entirely, keep D and N as bare math	Loses a pedagogically useful hook for no structural reason, since D does earn its name under the paper's own rule. Overcorrects.
Define S_{atom}/N as a new, named quantity	Mathematically available and not yet claimed by anything, but not yet shown to earn a name the way $\mathcal{E}_{\text{atom}}$ or D did. Premature until that scrutiny is done.
Reuse "density" for any of the above	Not recommended under any reading; the word is already spent and reusing it does not change that.
Pursue inertia in Assembly via any density-flavored metric	Closed, not live; already tested, already failed, with a specific number on file.

What stands up, by this accounting: keep N/D as mass-number/atomic-number, exactly as currently merged, on the strength of the earned, not evocative, identity behind D . Leave "density" retired. Leave Assembly's motion equation untouched, and treat inertia there as answered rather than open. The one genuinely undecided item is whether S_{atom}/N is worth naming at all; park it rather than name it, pending the same kind of scrutiny $\mathcal{E}_{\text{atom}}$ and D already passed and $S_{\text{atom}}/\mathcal{V}_{\text{atom}}/\gamma_{\text{atom}}$ already failed.